

The Riemann Hypothesis via Warped Stationarity

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Abstract

Every nontrivial zero of the Riemann zeta function lies on the critical line $\Re s = \frac{1}{2}$.

1 Introduction

Let Z be the Hardy Z -function. The warp $\Theta(t) = \vartheta(t) + ct$ with $c > -\inf \vartheta'$ produces $Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$, which is wide-sense stationary in the Wiener time-average sense with Bochner spectral measure supported in $[-1, 1]$. Paley–Wiener gives Y as entire of exponential type ≤ 1 , and Akhiezer–Laguerre–Pólya places $Y \in \mathcal{LP}$, so every zero of Y is real. The real-line factorization $Z = \sqrt{\Theta'} \cdot (Y \circ \Theta)$ together with the Hardy Z -function axiom gives $\Re \rho = \frac{1}{2}$.

2 Axioms

Axiom 1 (Hardy Z -function). $Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, $\vartheta(t) := \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Z is real on $\mathbb{R}$, $|Z(t)| = |\zeta(\frac{1}{2} + it)|$, and for $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$. Moreover the nontrivial zeros of ζ are exactly $\{\frac{1}{2} + it : t \in \mathbb{R}, Z(t) = 0\} \cup \{\rho : \Re \rho \neq \frac{1}{2}\}$.

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2}) \rightarrow +\infty$.

Axiom 3 (Hardy–Littlewood). $\int_0^T |\zeta(\frac{1}{2} + it)|^2 dt = T \log(T/(2\pi)) + (2\gamma_e - 1)T + O(T^{1/2} \log T)$.

Axiom 4 (Riemann–Siegel). For $t > 0$, $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$, $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$, $R(t) = O(t^{-1/4})$, $(d/dt) \arg R(t) = O(t^{-1/2})$.

Axiom 5 (Wiener Thm. 29). If $f \in L^2_{\text{loc}}(\mathbb{R})$ has time-average covariance $c_f(\eta)$ existing, positive-definite, continuous at 0, then

$$\mu[f](E) = \lim_{T \rightarrow \infty} \frac{2\pi}{T} \int_E \left| \frac{1}{2\pi} \int_0^T f(u) e^{-imu} du \right|^2 dm$$

for every bounded Borel $E \subset \mathbb{R}$.

Axiom 6 (Paley–Wiener for WSS). If $f \in L^2_{\text{loc}}(\mathbb{R})$ has $\text{supp } \mu[f] \subseteq [-\tau, \tau]$, then f extends to an entire function on $\mathbb{C}$ of exponential type $\leq \tau$ with $|f(z)| \leq \sqrt{\mu[f](\mathbb{R})} e^{\tau|\Im z|}$.

Axiom 7 (Akhiezer–Laguerre–Pólya). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type $\leq \tau$, real on $\mathbb{R}$, with autocorrelation measure $\mu[F]$ finite, non-negative, supported in $[-2\tau, 2\tau]$, then $F \in \mathcal{LP}$ and every zero of F is real.

3 Definitions

Definition 1. $\Theta(t) := \vartheta(t) + ct$, $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$.

Definition 2. $Y : \mathbb{R} \rightarrow \mathbb{R}$, $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$.

Definition 3. $K[T](\mu) := \frac{1}{2\pi} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$.

4 Lemmas

Lemma 4. $c_* \in (0, \infty)$, and for $c > c_*$, $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\Theta' > 0$.

Lemma 5. $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ for every $t \in \mathbb{R}$.

Lemma 6. $\vartheta''(t) > 0$ for $t > 0$.

Lemma 7. For $n \geq 1$, $t \geq 2\pi n^2$, $\omega_n(t) := (\vartheta'(t) - \log n)/(\vartheta'(t) + c) \in [0, 1)$.

5 Main Results

Theorem 8 (Time-average WSS of Y). $c[Y](\eta) := \lim_{U \rightarrow \infty} U^{-1} \int_0^U Y(u)Y(u + \eta) du$ exists for every $\eta \in \mathbb{R}$, is positive-definite, continuous at 0, with $c[Y](0) = 2$.

Proof. Change of variable $u = \Theta(t)$: $\int_0^U Y^2 du = \int_0^{\Theta^{-1}(U)} |Z|^2 dt$. The Hardy–Littlewood axiom together with $\Theta(T) \sim (T/2) \log(T/(2\pi))$ gives $c[Y](0) = 2$. For $\eta \neq 0$, insert the Riemann–Siegel expansion into $Z(t)Z(t + \delta_\eta(t))$, $\delta_\eta(t) = \eta/\Theta'(\tau)$, and split:

Diagonal ($m = n$, $\sigma = \sigma'$): contributes $(1/U) \sum_{n \leq N(t)} (1/n) \int \cos(\eta \omega_n(t)) dt$; bounded convergence via the frequency-ratio lemma gives a finite limit $A(\eta)$.

Antidiagonal ($m = n$, $\sigma = -\sigma'$): phase slope $|2\vartheta'(t)| - O(1) \rightarrow \infty$; one integration by parts yields $o(U)$.

Off-diagonal ($m \neq n$): phase slope $\geq \frac{1}{2} |\log(m/n)|$; one integration by parts with the Montgomery–Vaughan near-diagonal bound yields $o(U)$.

Remainder: $\int_0^T R^2 = O(T^{1/2})$ by Riemann–Siegel, so Cauchy–Schwarz against the main sum yields $O(T^{3/4}(\log T)^{1/2}) = o(U)$.

Positive-definiteness is inherited as a Cesàro limit of positive-definite kernels. Continuity at 0 follows from $|c[Y](\eta) - c[Y](0)|^2 \leq c[Y](0)(c[Y](0) - \Re c[Y](\eta))$ and continuity of A at 0. $\square$

Corollary 9. There exists a finite non-negative Borel measure $\mu[Y]$ on $\mathbb{R}$ with $c[Y](\eta) = \int e^{i\eta\beta} d\mu[Y](\beta)$ and $\mu[Y](\mathbb{R}) = 2$.

Theorem 10 (Band-limit). For $|\mu| > 1$, $\lim_{T \rightarrow \infty} 2\pi |K[T](\mu)|^2 / \Theta(T) = 0$.

Proof. Insert Riemann–Siegel into $K[T](\mu)$ and substitute $u = \Theta(t)$. The (n, σ) -mode gives $\mathcal{J}_{n,\sigma}(T, \mu) = \int_{u_n}^{\Theta(T)} G_{n,\sigma}(u) e^{iP(u)} du$, $u_n = \Theta(2\pi n^2)$, $G_{n,\sigma}(u) = x^{-1/2}$, $x = \Theta'(\Theta^{-1}(u))$, $P'(u) = \sigma \omega_n(\Theta^{-1}(u)) - \mu$. The frequency-ratio lemma gives $|P'| \geq |\mu| - 1 =: \lambda > 0$. With $Q = G/P' = x^{1/2}/\hat{P}(x)$, $\hat{P}(x) = (\sigma - \mu)x - \sigma(c + \log n)$, one integration by parts and $|\hat{P}(x)| \geq \lambda x$ give $|\mathcal{J}_{n,\sigma}(T, \mu)| \leq C_1(c, |\mu|) \beta_n^{-1/2}$, $\beta_n \geq c + \log n$. The remainder contributes $O(T^{-1/4})$. Summing over $n \leq N(T)$:

$$|K[T](\mu)| \leq C_1 \sum_{n=1}^{N(T)} \frac{1}{\sqrt{n(c + \log n)}} + O(T^{-1/4}) = O\left(\frac{T^{1/4}}{\sqrt{\log T}}\right).$$

Since $\Theta(T) \sim (T/2) \log T$, $2\pi |K[T](\mu)|^2 / \Theta(T) = O(T^{-1/2}/(\log T)^2) \rightarrow 0$. $\square$

Corollary 11. $\text{supp } \mu[Y] \subseteq [-1, 1]$.

Proof. Wiener Theorem 29 plus the band-limit theorem plus dominated convergence. $\square$

Theorem 12. Y extends to an entire function on $\mathbb{C}$ of exponential type ≤ 1 with $|Y(z)| \leq \sqrt{2} e^{|\Im z|}$.

Proof. Paley–Wiener axiom with $\tau = 1$, $\mu[Y](\mathbb{R}) = 2$. $\square$

Theorem 13. *Every zero of Y in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Y entire of exponential type ≤ 1 , real on $\mathbb{R}$, $\text{supp } \mu[Y] \subseteq [-1, 1] \subseteq [-2, 2]$; Akhiezer–Laguerre–Pólya with $\tau = 1$. $\square$

Theorem 14 (Riemann Hypothesis). *Every nontrivial zero of ζ satisfies $\Re \rho = \frac{1}{2}$.*

Proof. By the Hardy Z -function axiom, $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ for $t \in \mathbb{R}$. By the factorization lemma, $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$, so $Z(t) = 0 \iff Y(\Theta(t)) = 0$. By the preceding theorem, every zero of Y in $\mathbb{C}$ is real, so $Y(\Theta(t)) = 0$ for some $t \in \mathbb{R}$ iff $\Theta(t) \in Y^{-1}(0) \subset \mathbb{R}$, which is automatic since $\Theta(t) \in \mathbb{R}$. Therefore the nontrivial zeros of ζ lie on $\frac{1}{2} + i\mathbb{R}$, i.e. $\Re \rho = \frac{1}{2}$. $\square$